

Introduction to Artificial Intelligence (AI) – 2021

Assignment 2

Weight: 10%

Due Date: 23:59, 10 June 2021

This is an individual assignment.

Fire incident is a disaster that can potentially cause the loss of life, property damage and permanent disability to the affected victim. They can also suffer from prolonged psychological and trauma. Fire fighters are primarily tasked to handle fire incidents, but they are often exposed to higher risks when extinguishing fire, especially in hazardous environments. As seen recently in Cape Town; fire destroyed historical sites and buildings such as Rhodes Memorial and UCT's Jagger Library and wildlife around Table Mountain. Many people had to evacuate their residential buildings and some fire fighters were badly injured.

1. Conduct research and write a paper on **How Artificial Intelligence, ROBOTICS in particular can be used to detect and fight fire.**
2. Take a look at the next 10 years of AI and robotics and identify some of the new trends that are likely to emerge in this area – i.e. fire detection and fire fighting.
3. Your paper must be written in Microsoft Word using Times New Roman, font size 12, 1.5 spacing and a maximum of 5 pages excluding your list of references and cover page.
4. Use reputable and scholarly sources that is reputable websites/papers, academic websites and journals. Properly cite your sources, both in-text and in your list of references using Harvard Referencing.

Marking Rubric						
Criteria	Outstanding	Very Good	Good	Satisfactory	Unacceptable	Mark
	9-10	7-8	5-6	3-4	1-2	

Appropriate details	Judicious choice of details maximizes interest and understanding	All relevant details presented, but details not critical to understanding omitted	Enough critical details presented for understanding, unnecessary details generally omitted	Most important details included but may include too much or too little detail for easy understanding	Some critical details missing, unnecessary details may be present	*3 = 30
Clarity of explanations	Sophisticated use of language maximizes interest, enjoyment and comprehension; explanations very clear, factually correct	All explanations clear and easy to understand, factually correct	All explanations clear and easy to understand, factually correct	Overall meaning is understandable; possibly some areas of slight confusion or minor factual errors	Serious difficulty explaining ideas, major factual errors; lack of comprehensibility	* 2 = 20
Use of terminology	Correct use of all terminology, attention to nuances of meaning, judicious use of clearly defined jargon	All technical terms used correctly and defined clearly, including terms with different common meanings; overuse of jargon avoided	Few errors in use of terminology; definitions provided for technical terms, overuse of jargon avoided	Most terms used correctly, possibly some incorrect usage or use of unnecessary or undefined jargon	Jargon terms used incorrectly, without definition; attempting to sound "scientific" without understanding meaning of terms	*2 =20

Writing style	Sophisticated, elegant style, complex yet lucid sentence structure, flawless grammar	Error-free, easy to read writing style, well-practiced and polished use of language	Good basic writing style, easy to read, few errors, almost entirely in author's own words, little paraphrasing or unnecessary quotation	Mostly basic, correct writing style, relatively few errors and little awkwardness, minimal use of unnecessary quotation or paraphrasing	Serious errors and awkwardness, excessive use of quotation in place of author's own words, excessive paraphrasing	*1 =10
Organization	Develops persuasive arguments and explanations; effortless and seamless logical flow. Excellent and original synthesis of sources.	Well-chosen arguments and explanations; logical flow and connections always easy to follow. Sources used to prove points not just summaries.	Relevant arguments and explanations presented; builds case with mostly logically connected arguments, material from sources mostly integrated	Some arguments and explanations presented, possibly incomplete, or awkwardness in logical flow. Sources somewhat but not fully integrated.	Major sections missing or lack of logical flow; sources summarized but not connected.	*1 =10
Referencing	The paper follows a referencing style that complies with the Harvard style, and the in-text citations are made purposefully	The paper follows a referencing style that complies with the Harvard style, and the in-text citations are mostly purposeful.	The paper follows a referencing style that mostly complies with the Harvard style, though the in-text citations are not made purposefully.	The paper follows a referencing style that does not comply with the Harvard style, or only includes either the in-text citations or the reference list.	The paper does not include any citations and/or a reference list.	*1 =10
Total Mark						100

